DO NOW — RECALL

Database and SQL retrieval challenge

Name: _____ Class: _____ Date: _____

1. What is a primary key?

2. What does INNER JOIN return?

3. What is a foreign key?

4. A 3×2 zero-indexed array: specify row index range, column range and cell count.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Uniquely identifies a row.

2. Model answer

Matched records according to join condition.

3. Model answer

Column referencing a key in a related table.

4. Model answer

Rows 0–2, columns 0–1, total six cells.

Vocabulary — know these terms

primary key — Unique row identifier.

foreign key — Related table key reference.

INNER JOIN — Returns matching joined rows.

WHERE — Filters query results.

NEW LEARNING

Read, annotate and explain

Understand

Use a primary key for each Student row and a matching foreign key in Score.

Trace

INNER JOIN produces one result for each matching StudentID; Cleo has no matching Score and is absent.

Evaluate

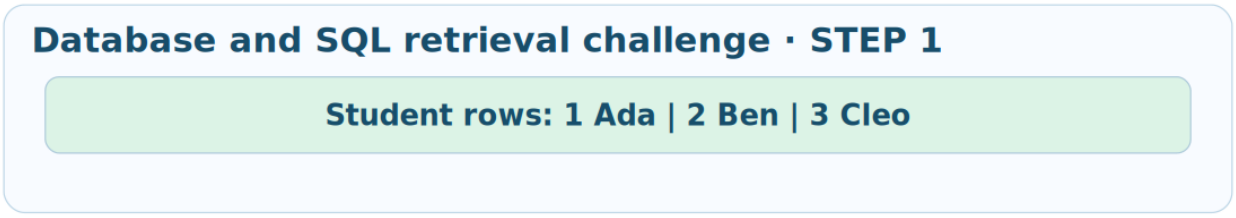
A WHERE clause can filter matched records; forgetting the join condition can produce unwanted combinations.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

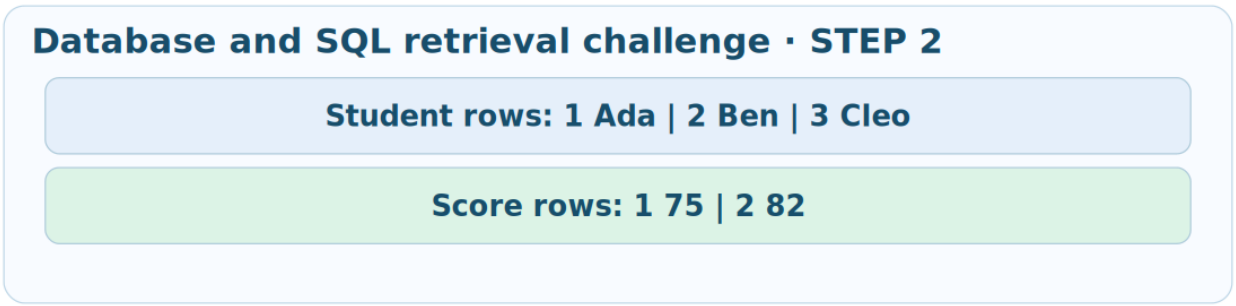
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



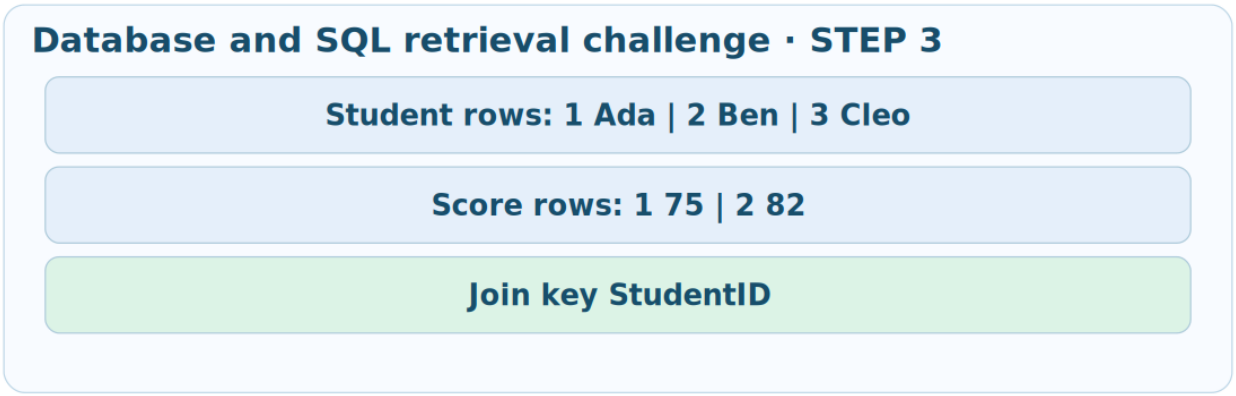
Database and SQL retrieval challenge · STEP 1

New Learning



Database and SQL retrieval challenge · STEP 2

New Learning

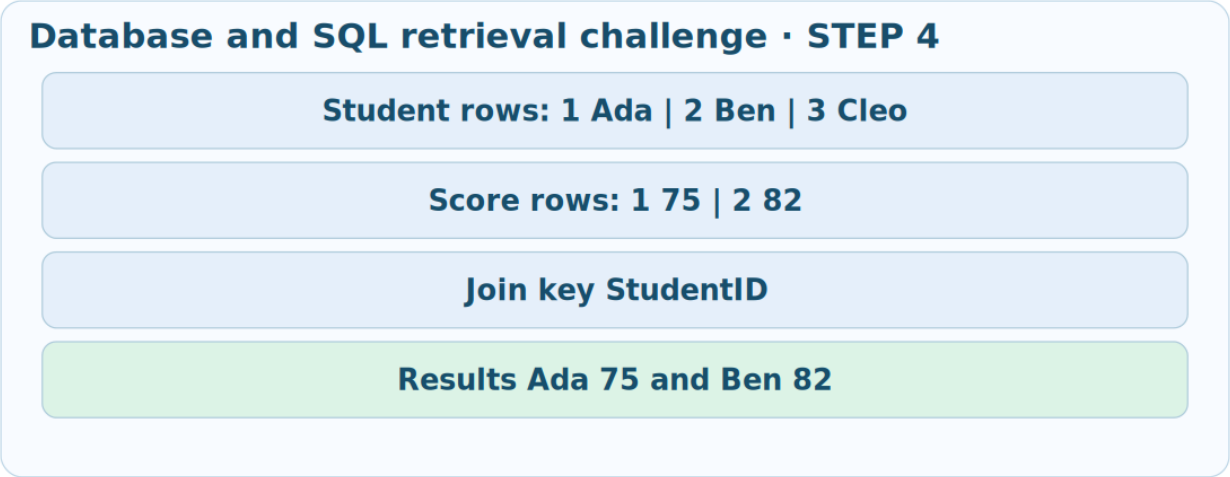


Database and SQL retrieval challenge · STEP 3

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Database and SQL retrieval challenge · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

Match Tutor and Class records on TutorID.

Predict first

Tutor 1 matches two class rows

Model and check

Tutor 2 has no class and is excluded from INNER JOIN.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Match Tutor and Class records on TutorID.

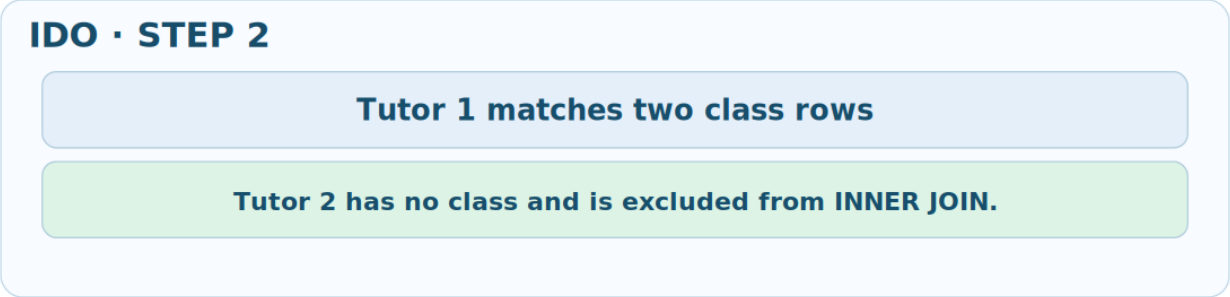
Tutor 1 matches two class rows



IDO · STEP 1

Match Tutor and Class records on TutorID.

Tutor 1 matches two class rows



IDO · STEP 2

WE DO — GUIDED PRACTICE

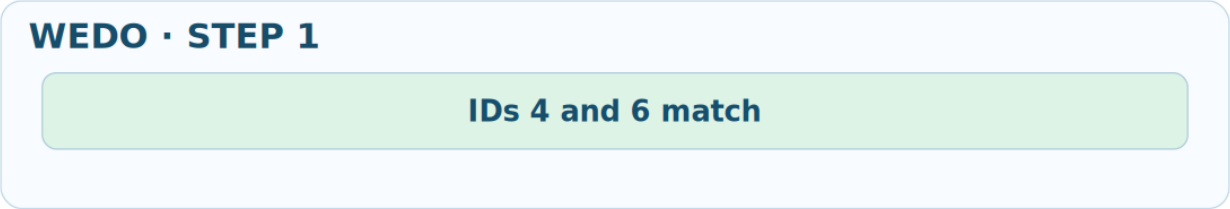
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Student IDs 4,5,6 but Mark IDs 4,6. What appears after INNER JOIN?

IDs 4 and 6 match



WEDO · STEP 1

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first IDs 4 and 6 match

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define primary key, foreign key, INNER JOIN and WHERE. [4]

Question 2

Write an INNER JOIN for Student and Score above selecting Name and Mark; give output. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why omitting a join condition may produce misleading results. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

INNER JOIN automatically includes rows without a key match.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

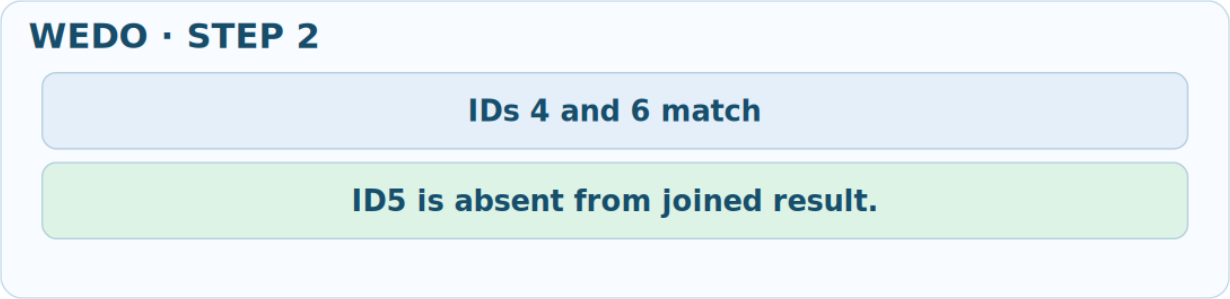
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Student IDs 4,5,6 but Mark IDs 4,6. What appears after INNER JOIN?

IDs 4 and 6 match



WEDO · STEP 2

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

ID5 is absent from joined result.

IMPROVE: Explain one correction to your own reasoning.